TruthTrack Visual Mockup Guide

Complete visual walkthrough of all TruthTrack mobile app screens with descriptions and user flows.

📱 Screen Overview

TruthTrack consists of 8 main screens organized into a bottom tab navigation with 5 tabs:

1. **Home** - Daily nutrition summary
2. **Log Meal** - Food logging with AI analysis
3. **Scanner** - Barcode scanning
4. **Summary** - Historical data and trends
5. **Profile** - Settings and user preferences

🔐 Authentication Flow

Login Screen

Purpose: First-time user authentication

Key Elements:

- TruthTrack title with tagline "Verified Nutrition Scanner"
- 4 feature icons highlighting core benefits:
  - 📸 AI Food Recognition
  - ✓ Verified Nutrition Data
  - 🔍 Transparent Audit Trail
  - 📊 Daily Tracking
- Email input field
- Optional name field
- "Sign In with Email" button (blue, primary action)
- Privacy notice at bottom

Design Notes:

- Clean, minimalist layout with soft blue (#0EA5E9) accents
- White background with subtle gray borders

- No clutter, focuses on trust and clarity
- Medical-grade aesthetic

User Flow:

1. User enters email address
2. Optionally enters name
3. Clicks "Sign In with Email"
4. Redirected to Home screen after authentication

🏠 Home Screen

Purpose: Daily nutrition summary and quick actions

Key Elements:

- Header: "Today's Nutrition" with current date
- Large calorie circle showing:
  - Current calories (e.g., 1,850)
  - Daily target (e.g., 2,000)
  - Percentage of target
- Progress bar indicating calorie consumption
- 4 macro cards showing:
  - Protein: 90g / 120g
  - Carbs: 210g / 250g
  - Fat: 60g / 70g
  - Fiber: 28g / 30g
- 2 quick action buttons:
  - 📸 Log Meal (opens camera/gallery)
  - 📱 Scan Barcode (opens scanner)
- "Today's Meals" section with meal cards
- Each meal card displays:
  - Time (e.g., "8:00 AM")
  - Meal type (Breakfast, Lunch, Dinner, Snack)
  - Confidence score circle (green/amber/red)

    - Ingredient list
    - Calorie and macro breakdown

Design Notes:

- Soft blues and greens for positive reinforcement
- No gamification elements
- Data-focused layout
- Responsive to different screen sizes

User Flow:

1. User opens app → Home screen loads
2. Sees daily summary at a glance
3. Can tap "Log Meal" to add new meal
4. Can tap "Scan Barcode" for packaged foods
5. Can tap any meal card to see details

📸 Meal Analysis Screen (Tier 1-3 Validation)

Purpose: AI food recognition with confidence scoring

Key Elements:

- Food image at top
- "AI Analysis" section showing identified ingredients:
    - Grilled Chicken 200g
    - White Rice 150g
    - Olive Oil 15ml
- Confidence score circle (0-100%) with color coding:
    - Green (⩾85%): High confidence
    - Blue (70-84%): Good confidence
    - Amber (50-69%): Moderate confidence
    - Red (<50%): Low confidence
- **"Help Me Verify" popup** (triggers when score < 85%):
    - Question 1: "Was this a small, medium, or large portion?"
        - 3 buttons: Small, Medium (selected), Large

    - Question 2: "Any dressing or sauce added?"
        - 2 buttons: Yes, No
- Nutrition breakdown:
    - Calories: 450 kcal
    - Protein: 35g
    - Carbs: 45g
    - Fat: 12g

Design Notes:

- Large food image for context
- Color-coded confidence indicator
- Simple, clear questions for verification
- Transparent about AI uncertainty

User Flow:

1. User takes photo or selects from gallery
2. AI analyzes image (Tier 1: Visual recognition)
3. Shows identified ingredients and confidence score
4. If score < 85%, "Help Me Verify" popup appears
5. User answers 2 clarifying questions
6. Confidence score updates based on answers
7. User can proceed to Tier 2 (Context Modifiers)

🎯 Context Modifiers Screen (Tier 2 Validation)

Purpose: Refine AI estimates with user context

Key Elements:

- Food image at top
- Identified food name: "Grilled Chicken Breast"
- "Refine Your Meal" section with toggle buttons:
    - Cooked in Oil (ON/OFF)
    - Restaurant Portion (ON/OFF)
    - Homemade (ON/OFF)

    - Add Dressing (ON/OFF)
    - Extra Sauce (ON/OFF)
    - Fried (ON/OFF)
- Each toggle shows current state (green for ON, gray for OFF)
- Updated nutrition display:
    - "Updated: 520 kcal, 35g protein, 45g carbs, 18g fat"
- Confidence score updates (e.g., "84%")

Design Notes:

- One-tap toggles for quick adjustments
- Real-time nutrition updates
- Confidence score increases with user input
- Grid layout for easy scanning

User Flow:

1. User sees identified food from Tier 1
2. Toggles relevant modifiers (e.g., "Cooked in Oil")
3. Nutrition values update in real-time
4. Confidence score increases
5. User can proceed to save or edit individual ingredients

🔍 Barcode Scanner Screen

Purpose: Scan packaged foods for verified nutrition data

Key Elements:

- Camera viewfinder at top
- Barcode scanning frame in center
- "Point camera at barcode" instruction text
- After scan, shows:
    - Product image
    - Product name (e.g., "Chicken Breast - Organic Farm")
    - "Nutrition Facts" label with ✓ "Verified" badge
    - Nutrition breakdown:

        - 450 kcal
        - 52g Protein
        - 0g Carbs
        - 25g Fat
    - "Add to Meal" button (blue)

Design Notes:

- Clean camera interface
- Clear scanning instructions
- Verified badge for label-verified data
- Instant lookup from Nutritionix database

User Flow:

1. User taps "Scan Barcode" from Home
2. Camera opens with scanning frame
3. User points camera at barcode
4. Barcode scanned automatically
5. Product details appear with verified nutrition
6. User taps "Add to Meal" to log

📋 Audit View Screen (Transparent Breakdown)

Purpose: Show exactly how nutrition was calculated

Key Elements:

- Meal header with:
    - Image thumbnail
    - Time (e.g., "Lunch – 12:30 PM")
- "Meal Breakdown" section with editable rows:
    - Grilled Chicken | 200g | 264 kcal (edit icon)
    - White Rice | 150g | 130 kcal (edit icon)
    - Olive Oil | 15ml | 120 kcal (edit icon)
- Total row:
    - "Total | 450 kcal | 35g protein | 45g carbs | 12g fat"

- Confidence Score section:
  - "Confidence Score: 78%"
  - Explanation: "Calculated based on ingredient recognition, portion estimation, and user inputs. Higher scores indicate more tracking accuracy."

Design Notes:

- Data-focused, no visual clutter
- Each ingredient is editable
- Shows exact calculation logic
- Transparent about confidence methodology

User Flow:

1. User taps meal card from Home
2. Audit View opens showing breakdown
3. User can tap edit icon to modify ingredient
4. Changes update total nutrition
5. Confidence score may adjust based on edits

📊 Daily Summary Screen (Historical Data)

Purpose: View nutrition data for any past date

Key Elements:

- Date selector at top with left/right arrows
- Selected date: "April 18, 2026"
- Large calorie circle:
  - 1,850 kcal
  - Goal: 2,000 kcal (92%)
- Progress bar showing consumption
- Macronutrients section:
  - Protein: 125g / 150g (83%)
  - Carbs: 220g / 250g (88%)
  - Fat: 65g / 75g (87%)
  - Each with horizontal progress bar

- "Meals Logged (3)" section:
  - Mini cards showing:
    - Breakfast 7:30 AM - 450 kcal
    - Lunch 12:15 PM - 680 kcal
    - Dinner 6:45 PM - 720 kcal
- "Trends (Last 7 Days)" section:
  - Bar chart showing calorie intake for past week
  - Days labeled: S, M, T, W, T, F, S

Design Notes:

- Easy date navigation
- Visual progress bars for quick scanning
- 7-day trend chart for pattern recognition
- No gamification, just data

User Flow:

1. User taps "Summary" tab
2. Current date summary loads
3. User can navigate to past dates with arrows
4. Sees macro breakdown and meal list
5. Can tap any meal to see details
6. Views 7-day trend chart

⚙️ Settings Screen (Profile & Preferences)

Purpose: User preferences, nutrition targets, and privacy

Key Elements:

- **Profile Section:**
  - User avatar
  - Name: "John Doe"
  - Email: "john@example.com"
- **Nutrition Targets Section:**
  - Daily Calorie Target: 2000 kcal (editable)

    - Protein Target: 150g (editable)
    - Carbs Target: 250g (editable)
    - Fat Target: 75g (editable)
- **Reminders Section:**
    - Breakfast Reminder: 7:00 AM (toggle ON)
    - Lunch Reminder: 12:30 PM (toggle ON)
    - Dinner Reminder: 6:30 PM (toggle ON)
- **Privacy & Data Section:**
    - Red "Delete My Data" button with warning icon
    - Permanently erases all user data
- **About Section:**
    - Version: 1.2.3

Design Notes:

- Clean, organized sections
- Editable fields for targets
- Toggle switches for reminders
- Red button for destructive action (Delete)
- Privacy-first approach

User Flow:

1. User taps "Profile" tab
2. Settings screen opens
3. Can edit nutrition targets
4. Can toggle reminders on/off
5. Can view app version
6. Can delete all data (with confirmation)

🎨 Design System

Color Palette

- **Primary Blue**: #0EA5E9 (soft, trustworthy)

- **Secondary Green**: #10B981 (positive, healthy)
- **White**: #FFFFFF (clean, minimal)
- **Background**: #F9FAFB (off-white, easy on eyes)
- **Text**: #1F2937 (dark gray, readable)
- **Borders**: #E5E7EB (light gray, subtle)

Confidence Score Colors

- **Green** (⩾85%): High confidence ✓
- **Blue** (70-84%): Good confidence
- **Amber** (50-69%): Moderate confidence ⚠
- **Red** (<50%): Low confidence ✗

Typography

- **Large Titles**: 30px, Bold
- **Section Headers**: 20px, Semibold
- **Body Text**: 16px, Regular
- **Small Text**: 12px, Regular
- **Font Family**: System font (SF Pro Display on iOS, Roboto on Android)

Spacing

- **Extra Small**: 4px
- **Small**: 8px
- **Medium**: 12px
- **Large**: 16px
- **Extra Large**: 20-24px

Border Radius

- **Small**: 4px
- **Medium**: 8px
- **Large**: 12px
- **Full**: 9999px (circles)

🔄 User Journeys

Journey 1: Log a Meal (Complete Flow)

1. Home Screen → Tap "Log Meal"
2. Camera opens → Take photo of meal
3. Meal Analysis Screen → AI identifies ingredients
4. Help Me Verify popup → Answer 2 questions (if score < 85%)
5. Context Modifiers Screen → Toggle relevant modifiers
6. Confirm & Save → Meal logged
7. Home Screen → Updated with new meal

Journey 2: Scan a Packaged Food

1. Home Screen → Tap "Scan Barcode"
2. Scanner Screen → Point at barcode
3. Product Details → Show verified nutrition
4. Tap "Add to Meal" → Food added to daily total
5. Home Screen → Updated with new item

Journey 3: Review Meal Details

1. Home Screen → Tap any meal card
2. Audit View Screen → See ingredient breakdown
3. Edit ingredient → Tap edit icon
4. Update quantity → Nutrition updates
5. Back to Home → Changes saved

Journey 4: Check Historical Data

1. Home Screen → Tap "Summary" tab
2. Daily Summary Screen → Current date shown
3. Navigate dates → Tap left/right arrows
4. View trends → See 7-day chart
5. Tap meal → See details

Journey 5: Adjust Settings

1. Home Screen → Tap "Profile" tab
2. Settings Screen → View current preferences
3. Edit targets → Update calorie/macro goals
4. Toggle reminders → Enable/disable notifications
5. Save changes → Return to Home

📐 Responsive Design

All screens are optimized for:

- **iPhone**: 6.1" (standard) and 6.7" (plus)
- **Android**: 5.5" to 6.7" screens
- **Landscape orientation**: Supported with adjusted layouts
- **Tablet**: Optimized for iPad Pro and Android tablets

♿ Accessibility Features

- **Color Contrast**: All text meets WCAG AA standards
- **Font Sizes**: Minimum 12px for readability
- **Touch Targets**: All buttons ⩾44x44px for easy tapping
- **Screen Reader**: Full VoiceOver/TalkBack support
- **Haptic Feedback**: Subtle vibrations for confirmations

🚀 Next Steps

To see the app in action:

1. **Clone the repository**

   Bash

   ```bash
   cd truthtrack-mobile
   ```

2. **Install dependencies**

   Bash

```
npm install
cd backend && npm install && cd ..
```

3. **Start the backend**

```bash
cd backend
npm run dev
```

4. **Start the frontend**

```bash
npm start
```

5. **Open in simulator**
   - Press 'i' for iOS
   - Press 'a' for Android
   - Press 'w' for web preview

📝 Notes

- All screens follow the medical-grade minimalist aesthetic
- No gamification elements (no badges, streaks, or achievements)
- Every meal displays a confidence score without exception
- "Help Me Verify" popup triggers automatically when confidence < 85%
- Audit View shows exact calculation breakdown with editable ingredients
- Privacy-first design with "Delete My Data" button
- All data is encrypted at rest

🎯 Key Principles

1. **Transparency**: Users see exactly how nutrition is calculated
2. **Trust**: Confidence scores and verification system build credibility
3. **Simplicity**: Minimal UI, maximum clarity

4. **Accuracy**: Three-Tier Validation ensures data quality
5. **Privacy**: User data is protected and erasable
6. **Medical-Grade**: Professional aesthetic, no clutter